Semester VI **B. Tech CSE (AI)**

BCSAI 601: BIG DATA ANALYTICS

3L + 0T + 0P + 3C **MM 100**

Unit 1: Introduction to Big Data

Introduction to Big Data, Characteristics of Data, and Big Data ,Evolution of Big Data, Definition of Big Data, Challenges with big data, Why Big data? Complexity of Big Data, Big Data Processing Architectures, Big Data Technologies, Data Warehouse, Traditional Business Intelligence versus Big Data. State of Practice in Analytics, Key roles for New Big Data Ecosystems, Big Data Analytics, Introduction to big data analytics, Classification of Analytics, Challenges of Big Data, Importance of Big Data, Big Data Technologies, Data Science, Responsibilities, Soft state eventual consistency, Data Analytics Life Cycle.

Unit 2: Analytical Theory and Methods

Analytical Theory and Methods: Clustering and Associated Algorithms, Association Rules, Apriori Algorithm, Candidate Rules, Applications of Association Rules, Validation and Testing, Diagnostics, Regression, Linear Regression, Logistic Regression, Additional Regression Models

Unit 3: Advanced Analytics Technology And Tools

Analytics for Unstructured Data, Use Cases, MapReduce, Apache Hadoop, The Hadoop Ecosystem, Pig, Hive, Hbase, Mahouth, NoSQL, SQL Essentials Joins, Set Operations, Grouping Extensions, In-Database Text Analysis, Advanced SQL, Window Functions, Userdefined Functions and Aggregates, Ordered Aggregates, MADlib

Unit 4: Hadoop Distributed File System Architecture

HDFS Architecture, HDFS Concepts, Blocks NameNode, Secondary NameNode, DataNode, HDFS Federation, HDFS High Availability, Basic File System Operations, Data Flow, Anatomy of File Read, Anatomy of File Write, Anatomy of a MapReduce Job Run

Unit 5: Processing Your Data With Mapreduce

Getting to know MapReduce, MapReduce Execution Pipeline, Runtime Coordination and Task Management, MapReduce Application, Hadoop Word Count Implementation

Text /Reference Books

1. Data Warehousing in the Age of Big Data by Krish Krishnan, Morgan Kaufmann.
2. A.Ohri, "R for Business Analytics", Springer, 2012.
3. Big Data Analytics with R and Hadoop by Vignesh Prajapati
4. Principles of Big Data Preparing, Sharing, and Analyzing Complex Information, 1st Edition, by J Berman, published by Morgan Kaufmann
5. "Big Data Analytics - From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph" By David Loshin, Morgan Kaufmann